Research Report: Plasma Metabolites and Gait Performance

1. Executive Summary

This study investigated the association between 133 plasma metabolites and gait performance metrics in the HPP 10K cohort (N=380, Age 40-70). We found that while individual metabolite correlations are small (typically rho < 0.2), a multivariate metabolic profile can predict gait speed with a Pearson R of 0.227. The most significant associations were observed with sulfate-conjugated metabolites and specific lipid species.

2. Methodology

- Cohort: HPP 10K, filtered to ages 40-70, study_type == 10.
- Metrics: LStep_length_m, Cadence, Step_width_m, and Romberg sway.
- Analysis: Spearman correlations adjusted for age, gender, and BMI using the residuals method.
- Correction: Benjamini-Hochberg FDR correction applied across all pairs.
- Modeling: Nested Cross-Validation comparing Ridge and LightGBM regression.

3. Key Findings

The strongest adjusted association was found for:
- 2 5-Dihydroxybenzenesulfonic Acid _Hydroquinone sulfate_Pyrocatechol sulfate: rho_adj = 0.177 (p_adj = 0.001)
Target: LStep_length_m: median - TM3

Predictive modeling showed that gait speed is the most 'metabolically predictable' gait metric (R = 0.227), whereas stance time showed significantly lower predictability (R = 0.093).

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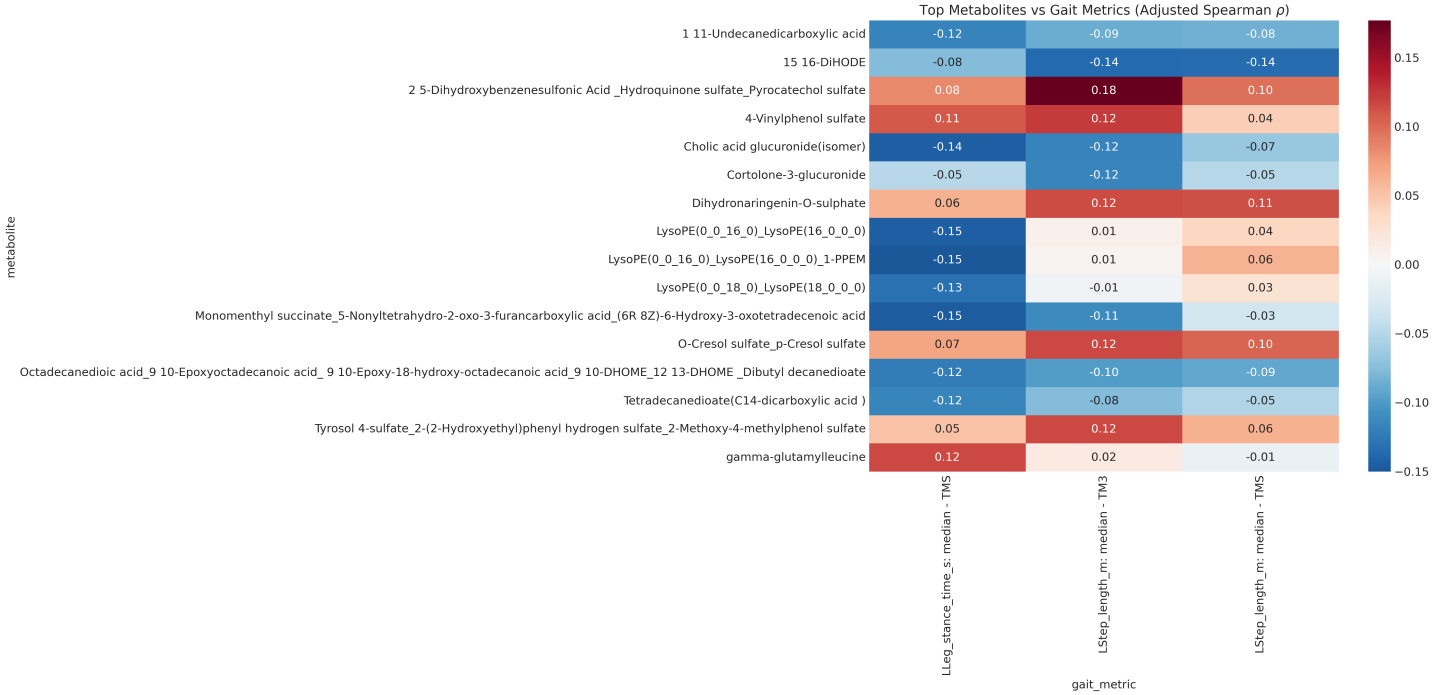

Figure 1: Top Metabolites vs Gait Metrics (Adjusted Spearman rho).

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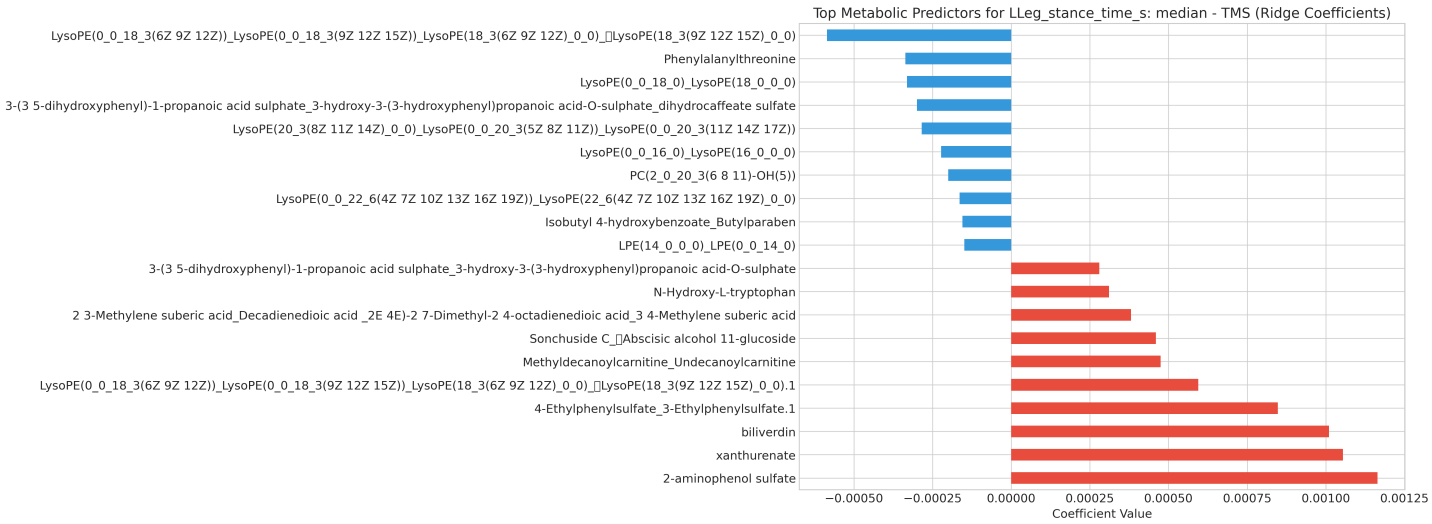

Figure 2: Top metabolic predictors for s: median - TMS.